

NAME(S):

Mathematical Recursion

Begin the following as paired-programming in class, but be sure to individually finish any problems you don't get to (hand in written answers of **0.25 engagement points**, check website for due date).

For each mathematical scenario below, write the base case and the recursive step

Write your answers as if you were defining a recursive function.

1. What is $n!$, n factorial?

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2. What is the sum of integers from 1 to n ?

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3. What is the n th Fibonacci number?

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4. What is the result of raising x to the power n , x^n ?

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5. What is the sum of the digits in a positive integer n ?

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6. What is the result of multiplying a positive integer n by another positive integer m ?

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